COER UNIVERSITY

ASSIGNMENT-2

Course: MCA | Subject:Machine Learning (MCA201T)

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Q1. Explain the concept of Machine Learning as a computational approach to learning from data. Discuss the relationship among data, features, learning algorithms, hypotheses/models, and predictions.

Machine Learning as a Computational Approach

Machine Learning (ML) is a computational approach in which a program does not follow hand-written rules for every case, but instead ***infers the rules from examples (data)***. The program is given experience in the form of data, and an algorithm search for a general pattern that explains that data. The learned pattern is then applied to new, unseen cases. According to Tom Mitchell, a program learns from experience E with respect to a task T and performance measure P if its performance at T, measured by P, improves with E.

Formally, in supervised learning we are given a training set of examples $\{(x_1, y_1), ..., (x_n, y_n)\}$, where x is the input (features) and y are the output. The learner searches a set of candidate functions called the ***hypothesis space H*** and selects a hypothesis h that maps x to y and minimises a loss (error) on the training data, in the hope that it will also perform well on new data. This ability to perform well on unseen data is called ***generalisation***, and it is the real goal of learning.

Key Components and Their Relationship

Component	***Meaning***	***Example (house price prediction)***
Data	Collection of observed examples (instances) from which the system learns; may be labelled or unlabelled.	Records of 10,000 houses sold earlier.
Features	Measurable attributes or properties of each instance, used as the input variables (the representation of data). Good features make patterns easy to learn.	Area, number of rooms, location, age of the house.
Target / label	The quantity to be predicted (present in supervised learning).	Selling price.
Learning algorithm	The procedure that searches the hypothesis space and adjusts parameters to minimise a loss function (e.g., gradient descent, tree-splitting).	Linear regression trained by least squares.
Hypothesis / model	The learned function with fixed parameters, which represents the knowledge extracted from data.	Price = 3000 x area + 50000 x rooms + ... (learned coefficients).

Component	*Meaning*	*Example (house price prediction)*
Prediction	The output produced by the model for a new input (inference).	Predicted price of a new house not seen before.

Relationship (flow): Data -> Feature extraction / selection -> Learning algorithm (searches hypothesis space, minimises loss) -> Hypothesis / trained model -> Predictions on new data -> Evaluation and feedback -> improved data / features / algorithm.

- The ***quality of data and features*** sets the upper limit of what can be learned; a good algorithm cannot fix irrelevant or misleading features.
- The ***algorithm and its hypothesis space*** decide what kind of patterns can be represented (a linear model can only represent linear relations; a neural network can represent very complex ones).
- The ***model*** is the result of training: the algorithm is applied to the data once, the model is then reused many times.
- ***Predictions*** are evaluated against reality, and this evaluation guides improvement of features, algorithm choice and data collection, so the whole process is iterative.

Q2. Compare supervised, unsupervised, and reinforcement learning using a structured discussion of training data, feedback mechanism, objective function, typical tasks, and representative algorithms.

Aspect	*Supervised Learning*	*Unsupervised Learning*	*Reinforcement Learning*
Training data	Labelled data: pairs of input and correct output (x, y).	Unlabelled data: only inputs x, no target.	No fixed dataset; experience (state, action, reward, next state) is generated by an agent interacting with an environment.
Feedback mechanism	Direct and immediate: error between prediction and true label acts as a teacher.	No external feedback; the algorithm judges itself using internal measures of structure.	Indirect and often delayed: a scalar reward or penalty after actions; no correct action is given.
Objective function	Minimise a loss between predicted and true values, e.g., mean squared error for regression, cross-entropy for classification.	Optimise a structural criterion, e.g., minimise within-cluster variance (k-means), maximise variance retained (PCA), minimise reconstruction error (autoencoder).	Maximise expected cumulative (discounted) reward, i.e., learn a policy that gives the highest long-term return.

Aspect	*Supervised Learning*	*Unsupervised Learning*	*Reinforcement Learning*
Typical tasks	Classification (spam detection), regression (price prediction).	Clustering (customer segmentation), dimensionality reduction, anomaly detection, association rule mining.	Sequential decision making: game playing, robot control, autonomous driving, resource allocation.
Representative algorithms	Linear / logistic regression, decision trees, random forest, SVM, k-NN, Naive Bayes, neural networks.	k-means, hierarchical clustering, DBSCAN, PCA, Apriori, Gaussian mixture models, autoencoders.	Q-learning, SARSA, Deep Q-Network, policy gradient, actor-critic, Monte Carlo methods.
Main challenge	Cost of obtaining labels; overfitting.	Difficult to evaluate; results may be ambiguous.	Exploration versus exploitation; need for many interactions; reward design.

Explanation with examples: In ***supervised learning***, a bank trains a model on past loan applications labelled as "repaid" or "defaulted" to predict default for new applicants. In ***unsupervised learning***, a retailer gives purchase histories to k-means without any labels and discovers groups such as "bargain hunters" and "premium buyers". In ***reinforcement learning***, a robot vacuum learns to clean a room efficiently by receiving positive reward for cleaned area and negative reward for collisions or battery drain.

Hybrid forms also exist: ***semi-supervised learning*** uses a few labelled and many unlabelled examples, and ***self-supervised learning*** creates labels from the data itself (as in language models).

Q3. Select any three application domains of Machine Learning and analyze how ML can improve decision-making in each domain. Also discuss at least one limitation or risk associated with each application.

Domain 1: Healthcare

- ***Applications:*** Detecting diseases from medical images (X-ray, MRI, retinal scans), predicting patient deterioration or readmission, and personalising treatment.
- ***Improvement in decision-making:*** ML analyses thousands of images and records quickly and consistently, helping doctors to identify high-risk patients earlier, prioritise urgent cases, reduce missed diagnoses and choose treatment based on evidence from similar patients. The doctor remains the final decision-maker, supported by data-driven insights.

Limitation / risk: A model trained on data from one hospital or population may perform poorly on others (data bias), and errors have life-and-death consequences. Deep models are hard to explain, which reduces doctors' trust, and patient data raises serious privacy concerns.

Domain 2: Finance and Banking

- ***Applications:*** Credit scoring and loan approval, fraud detection, algorithmic trading, risk assessment and customer churn prediction.

• ***Improvement in decision-making:*** ML uses many more variables than traditional scorecards, so it can estimate default risk more accurately, approve deserving borrowers faster, detect suspicious transactions in real time and reduce financial losses. Decisions become faster, more consistent and scalable.

• ***Limitation / risk:*** Models can inherit historical bias and unfairly reject applicants of certain groups. Fraudsters change tactics, so models decay (concept drift). Regulators also require explanations for credit decisions, which is hard for complex models.

Domain 3: Agriculture

- ***Applications:*** Crop yield prediction, plant disease detection from leaf images, soil and irrigation recommendations, pest outbreak forecasting and market price prediction.

• ***Improvement in decision-making:*** Using weather, soil sensor and satellite data, ML helps farmers decide when to sow, irrigate, fertilise and harvest, which increases yield, saves water and inputs, reduces crop losses and helps in planning sales.

• ***Limitation / risk:*** Models depend on good local data, which is often scarce in rural areas; predictions may fail under unusual weather or new pests. Farmers may lack access to technology or digital skills, so benefits can be unequal, and wrong advice can cause financial loss.

Domain	***How ML improves decisions***	***Key risk***
Healthcare	Earlier and more accurate diagnosis; risk-based prioritisation.	Bias, wrong diagnosis, privacy, lack of explainability.
Finance	Accurate credit / fraud decisions in real time.	Unfair discrimination, model drift, regulatory need for explanation.
Agriculture	Data-driven farming: better yield and resource use.	Poor local data, unreliable in unusual conditions, digital divide.

Q4. Design a conceptual Machine Learning system for detecting fraudulent online transactions. Describe the problem definition, data requirements, feature selection, model training, validation strategy, evaluation metrics, deployment, and monitoring stages.

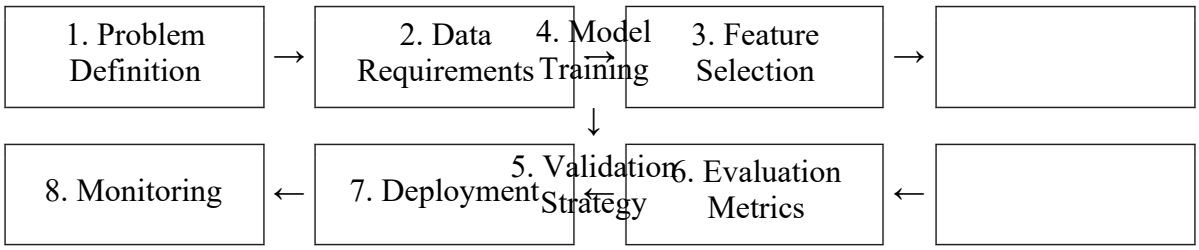


Figure: Conceptual pipeline of a fraud detection system

1. Problem Definition

Goal: to decide, within milliseconds, whether an online transaction (card, UPI, net banking, wallet) is fraudulent so that fraud can be blocked or verified while genuine customers are not disturbed. It is formulated as a ***supervised binary classification*** problem (fraud = 1, genuine = 0) that produces a fraud probability score, supported by unsupervised anomaly detection for new fraud types.

Success is defined in business terms: reduce fraud loss, keep false alarms low, and respond within about 100 milliseconds.

2. Data Requirements

- ***Transaction data:*** amount, timestamp, merchant and merchant category, payment method, currency, location.
- ***Customer and account data:*** account age, usual spending pattern, past transaction history, credit limit.
- ***Device and network data:*** device ID, IP address, browser, geolocation, number of failed login attempts.
- ***Labels:*** confirmed fraud (from customer complaints, chargebacks, investigation outcomes) and confirmed genuine cases. Labels arrive with a delay of days or weeks.
- The data must be large, recent and representative; it is highly ***imbalanced*** (fraud is well below 1%) and must be handled with privacy and security controls (masking, encryption, access control).

3. Feature Selection and Engineering

- ***Raw features:*** amount, hour of day, merchant category, country.
- ***Derived / behavioural features:*** number and total amount of transactions in the last 1 hour / 24 hours (velocity), difference from the customer's average spend, distance from the previous transaction location, time since last transaction, new device or new merchant flag, mismatch between billing and shipping address.
- ***Selection methods:*** remove irrelevant and highly correlated features, and rank features using mutual information, tree-based feature importance or recursive feature elimination. Categorical variables are encoded and numerical ones scaled where needed.
- Care is taken to avoid ***data leakage***: only information available at the time of the transaction may be used (for example, not the chargeback outcome).

4. Model Training

- Start with a simple, interpretable ***baseline*** (logistic regression), then try stronger models such as random forest and gradient boosting (XGBoost / LightGBM), and optionally neural networks. Anomaly detectors (isolation forest, autoencoders) complement supervised models.
- Handle class imbalance using class weights / cost-sensitive learning, undersampling of genuine cases or oversampling (SMOTE) applied ***only to the training data***.
- Tune hyperparameters using the validation data and select the model that gives the best trade-off of fraud catch rate versus false alarms.

5. Validation Strategy

- Use a ***time-based split***: train on older transactions and validate / test on newer ones, because the system will predict the future and fraud patterns change over time. Random shuffling would leak future information.
- Use stratified sampling or time-series (walk-forward) cross-validation so that fraud cases are present in every fold.

- Keep a final untouched ***test set***; apply resampling only inside training folds.
- Account for delayed labels by leaving a gap before the validation period.

6. Evaluation Metrics

- Accuracy is not used as the main metric because of imbalance.
- ***Recall*** (fraud detection rate), ***precision*** (share of alerts that are truly fraud), ***F1-score***, and ***PR-AUC*** (area under precision-recall curve); ROC-AUC as a supplementary measure. ***Cost-based metrics:*** total expected loss = (cost of missed fraud x false negatives) + (cost of review / customer inconvenience x false positives); the decision threshold is chosen to minimise this cost.
- Operational metrics: false positive rate, alert volume the review team can handle, and response latency.

7. Deployment

- The model is exposed as a low-latency ***real-time scoring service (API)*** attached to the payment flow, with a feature store supplying up-to-date behavioural features.
- Based on the score, action is taken: approve (low risk), ask for OTP / additional authentication (medium risk), send for manual review or decline (high risk). A rule engine for known fraud patterns is combined with the ML model.
- Safe rollout: first in ***shadow mode*** (scores are recorded but not acted upon), then A/B or gradual release, with version control and a rollback plan. Scalability and security of the service are ensured.

8. Monitoring and Maintenance

- Track input data drift (e.g., population stability index), score distribution, fraud rate, alert rate, false positive complaints and latency.
- Measure real performance as confirmed labels arrive (chargebacks and investigation results), and compare against thresholds.
- Retrain periodically or when drift is detected, since fraudsters adapt (concept drift and adversarial behaviour).
- Maintain feedback loop from analysts, audit logs and explanations (e.g., reason codes) for accountability and regulatory compliance.

Q5. Discuss the evolution of Machine Learning and identify the major factors responsible for the recent growth of AI and ML, such as large datasets, improved algorithms, specialized hardware, cloud computing, and increased computational capacity.

Evolution of Machine Learning

Period	***Development***
1940s-50s	McCulloch-Pitts artificial neuron (1943); Turing's "Computing Machinery and Intelligence" (1950); Arthur Samuel's self-learning checkers program and the term "machine learning" (1950s); Rosenblatt's Perceptron (1957); Dartmouth Conference (1956) starts AI as a field.

Period	*Development*
1960s-70s	Nearest-neighbour methods; Minsky and Papert showed limits of single-layer perceptrons (1969), causing reduced funding, the first "AI winter"; focus on symbolic AI and expert systems.
1980s	Backpropagation popularised (1986) enabling multi-layer neural networks; decision trees (ID3, CART); Q-learning for reinforcement learning; convolutional networks for digit recognition (LeCun).
1990s-2000s	Statistical machine learning: Support Vector Machines (1995), ensemble methods such as boosting and Random Forests, LSTM (1997), probabilistic models; internet data starts to grow. Deep belief networks (2006) revive deep learning.
2010s	ImageNet (2009) and AlexNet on GPUs (2012) start the deep learning revolution; GANs (2014), ResNet (2015), AlphaGo (2016); speech, vision and translation reach near-human levels.
2017 onwards	Transformers (2017), BERT and GPT models, large language models, diffusion models and multimodal foundation models; generative AI enters everyday use.

Major Factors Behind the Recent Growth

Factor	*Contribution*	*Example*
Large datasets (big data)	Internet, social media, smartphones, IoT sensors and digitisation produce huge amounts of data; modern models, especially deep networks, improve with more data.	ImageNet (millions of labelled images); web-scale text for language models.
Improved algorithms	New architectures and training techniques made deep models practical and stable: ReLU activation, dropout, batch normalisation, Adam optimiser, CNNs, LSTMs, attention and Transformers, transfer learning.	Transformer replaced RNNs in language tasks.
Specialised hardware	GPUs perform parallel matrix operations very fast; TPUs and AI accelerators are designed for tensor computation; training time dropped from months to days.	AlexNet trained on two GPUs (2012).
Cloud computing	Provides on-demand access to powerful hardware and storage without heavy investment, allowing start-ups and students to train large models; supports managed ML services and deployment at scale.	AWS, Google Cloud, Azure ML platforms.
Increased computational capacity	Advances in processors (Moore's law), memory, networks and distributed / parallel training allow models with billions of parameters to be trained.	Training of large language models on clusters of thousands of accelerators.

Factor	*Contribution*	*Example*
Open-source tools and community	Free frameworks (TensorFlow, PyTorch, scikit-learn), open datasets, pre-trained models and research sharing speeds up progress.	Pre-trained models fine-tuned for new tasks.
Investment and business demand	Industry and government funding and clear commercial value (recommendations, automation) drive research and adoption.	Use of ML in search, e-commerce, banking and healthcare.

Q6. Explain the common causes of failure in Machine Learning projects. Discuss how data quality, feature representation, inappropriate model selection, overfitting, bias, changing data distributions, and inadequate evaluation can affect system performance.

Cause	*Effect on system performance*	*Example*	*Prevention*
Poor data quality	Missing, wrong, duplicate, inconsistent or insufficient data leads to inaccurate and unreliable models ("garbage in, garbage out").	Wrongly entered ages in a medical dataset lead to wrong risk predictions.	Data cleaning, validation checks, more and better data collection.
Poor feature representation	Irrelevant, redundant or badly scaled/encoded features hide the real pattern; missing important features limits accuracy whatever the algorithm.	Predicting delivery time without distance or traffic features.	Domain knowledge, feature engineering and selection, proper encoding and scaling.
Inappropriate model selection	A model that is too simple underfits; too complex overfits; a model unsuitable for the data type or requirement (e.g., black box where explanation is needed) causes failure in practice.	Using linear regression for highly non-linear data.	Start with baselines, compare several models, consider interpretability, speed and data size.
Overfitting	Model memorises training data including noise and performs badly on new data; test performance is far lower than training performance.	100% training accuracy but 70% on new data.	More data, regularisation, cross-validation, early stopping, simpler models, dropout.
Bias (in data and model)	Unrepresentative or prejudiced data produces unfair or systematically wrong predictions for some groups.	Recruitment model favouring men because past hires were mostly men.	Representative sampling, bias audits, fairness metrics, diverse teams.

Cause	*Effect on system performance*	*Example*	*Prevention*
Changing data distributions (drift)	When real-world data changes after deployment, the model's learned patterns become outdated and accuracy falls.	Customer buying behaviour changed after a pandemic; fraud patterns change.	Monitoring, drift detection, regular retraining.
Inadequate evaluation	Wrong metrics, data leakage, no independent test set or unrealistic test data gives falsely high performance that fails in production.	99% accuracy on imbalanced fraud data with zero fraud detected.	Proper train / validation / test split, suitable metrics, time-based validation, real-world testing.

Other Common Causes of Failure

- ***Unclear problem definition or business goal*** and unrealistic expectations of ML.
- ***Lack of domain knowledge*** and poor communication between data scientists and stakeholders.
- ***Difficulty in deployment and integration*** with existing systems (many models never reach production); lack of MLOps and monitoring.
- ***Insufficient computing resources, time or skilled people***, and ignoring cost, latency, privacy and ethical requirements.

Q7. Data Science and Machine Learning are closely related but not identical disciplines. Compare them using a practical business problem and show which activities belong primarily to Data Science and which specifically involve Machine Learning.

Basic Comparison

Basis	*Data Science*	*Machine Learning*
Nature	Multidisciplinary field that extracts insights and knowledge from data using statistics, programming, domain knowledge and visualisation.	A subfield of AI focused on algorithms that learn from data to make predictions or decisions.
Goal	Understand data, answer business questions and support decisions.	Build accurate predictive / decision models that improve with data.
Scope	Entire data lifecycle: collection to communication.	Modelling stage: training, tuning, evaluation of models.
Output	Insights, reports, dashboards, recommendations, models.	Trained models / prediction systems.

Practical Business Problem: Customer Churn in a Telecom Company

A telecom company is losing customers and wants to understand why they leave and to identify customers who are likely to leave, so that it can offer them retention plans.

Activity	*Belongs to*	*What is done in the churn problem*
Business understanding and problem framing	Data Science	Define churn, the retention goal and success measures (e.g., reduce churn by 10%).
Data collection and integration	Data Science	Combine billing, usage, complaints, plan and customer-service data from databases using SQL / ETL.
Data cleaning and preparation	Data Science	Handle missing values, duplicates, wrong entries, merge tables.
Exploratory data analysis and visualisation	Data Science	Find that churn is higher among customers with many complaints or month-to-month contracts; create charts and dashboards.
Statistical analysis / hypothesis testing	Data Science	Test whether a price increase significantly affected churn; A/B testing of offers.
Feature engineering	Both	Create features such as average monthly usage, number of calls to support, tenure; select those useful for prediction.
Choosing and training a predictive model	Machine Learning	Train logistic regression, random forest or gradient boosting to predict the probability that a customer will churn.
Hyperparameter tuning and cross-validation	Machine Learning	Tune model settings; avoid overfitting.
Model evaluation	Machine Learning	Measure precision, recall, F1, ROC-AUC; handle class imbalance.
Deployment of the model	ML (with engineering)	Integrate the scoring model into the CRM system to flag high-risk customers every month.
Interpretation and storytelling	Data Science	Explain the main churn drivers to management and recommend actions (offers, service improvement); measure business impact.
Monitoring and retraining	Both	Track model accuracy and data drift; update the model and the analysis regularly.

Q8. Suppose an e-commerce company wants to recommend products to users. Determine whether supervised, unsupervised, or reinforcement learning or a combination would be appropriate. Justify your choice and describe the possible data and feedback involved. Choice: A Combination (Hybrid Approach)

No single paradigm is enough for a good recommender system. A ***combination*** of unsupervised, supervised and reinforcement learning is the most appropriate, with each used for the part of the problem it suits.

Learning type	*Use in recommendation*	*Justification*
Unsupervised	Clustering users with similar behaviour and items with similar properties; collaborative filtering based on similarity; learning embeddings (matrix factorisation, autoencoders); association rules for "frequently bought together".	Most users and products have no labels; patterns of similar users and items must be discovered from interaction data. Useful for candidate generation and for new-user segments.
Supervised	Predicting the probability that a user will click or buy a product (click-through / conversion prediction); predicting ratings; ranking candidate products (learning to rank) using models such as logistic regression, gradient boosting or neural networks.	Past clicks, purchases and ratings provide labelled examples (user, item, outcome). Supervised models rank the candidates precisely using many features.
Reinforcement (bandits / RL)	Choosing which products to show in a sequence over time, balancing exploration (trying new items) and exploitation (showing known favourites) to maximise long-term rewards such as purchases and customer lifetime value.	User preferences change and recommendations affect future behaviour; RL / contextual bandits optimise long-term outcome and learn online from user response.

Typical Architecture

1. ***Candidate generation*** (unsupervised / embeddings / collaborative filtering): narrow millions of products to a few hundred.
2. ***Ranking*** (supervised): score each candidate by probability of click or purchase using user, item and context features.
3. ***Re-ranking and exploration*** (bandit / RL): apply business rules, diversity, and exploration; learn from live feedback.

Data Involved

- ***User data:*** profile, location, device, past purchases, browsing history, search queries, wishlist, cart activity.
- ***Item data:*** category, brand, price, description, images, ratings, stock, seasonality.
- ***Interaction data:*** views, clicks, add-to-cart, purchases, returns, dwell time, ratings and reviews.
- ***Context data:*** time, day, festival / sale season, device type, current session.

Feedback

- ***Explicit feedback:*** star ratings, reviews, likes, "not interested" marks. It is accurate but scarce.
- ***Implicit feedback:*** clicks, time spent on a product, add-to-cart, purchase, repeat purchase, skipping. It is abundant but noisy (a click does not always mean liking).

- ***Reward signal for RL:*** e.g., +1 for click, higher reward for purchase, penalty for returns or leaving the site; delayed feedback such as long-term customer retention.

Challenges

Cold start (new users or products with no history), data sparsity, popularity bias and filter bubbles (users see only similar items), privacy of user data, and the need for real-time responses. Techniques such as content-based features, popularity-based fallback, exploration and diversity constraints help overcome them.

Q9. Explain the importance of selecting appropriate evaluation measures for Machine Learning systems. Why can accuracy be misleading for some problems? Illustrate your answer using an imbalanced classification scenario.

Importance of Appropriate Evaluation Measures

- Evaluation tells us ***how well the model will perform on unseen data*** and whether it is good enough for real use.
- Different problems have different ***costs of errors***; the metric must reflect the real objective (missing a cancer case is far worse than a false alarm).
- It allows ***fair comparison*** of models and guides tuning and model selection.
- A wrong metric may optimise the wrong behaviour and give ***false confidence***, resulting in failure after deployment.
- It helps to reveal problems such as overfitting, bias towards a class, and poor performance for subgroups.

Why Accuracy Can Be Misleading

Accuracy = (TP + TN) / (TP + TN + FP + FN), the fraction of all predictions that are correct. It counts every error equally and is dominated by the majority class, so on ***imbalanced datasets*** a model can achieve very high accuracy simply by always predicting the majority class, while completely failing on the minority class, which is normally the class of interest.

Illustration: Fraud Detection with Imbalanced Data

Suppose 1,000 transactions contain only 10 frauds (1%) and 990 genuine ones.

Model A (naive): predicts every transaction as genuine.

	Predicted Fraud	***Predicted Genuine***
Actual Fraud (10)	TP = 0	FN = 10
Actual Genuine (990)	FP = 0	TN = 990

Accuracy = (0 + 990) / 1000 = ***99%***, but recall = 0 / 10 = ***0%***: it detects no fraud at all and is useless, although it looks excellent.

Model B (real classifier):

	Predicted Fraud	***Predicted Genuine***
Actual Fraud (10)	TP = 7	FN = 3

	Predicted Fraud	***Predicted Genuine***
Actual Genuine (990)	FP = 20	TN = 970

- Accuracy = (7 + 970) / 1000 = ***97.7%*** (lower than Model A!)
- Precision = TP / (TP + FP) = 7 / 27 = ***25.9%***
- Recall (sensitivity) = TP / (TP + FN) = 7 / 10 = ***70%***
- F1-score = 2 x Precision x Recall / (Precision + Recall) = ***about 0.38***
- Specificity = TN / (TN + FP) = 970 / 990 = ***98%***

Although Model B has lower accuracy, it is far more useful because it catches 70% of the frauds.

Accuracy ranks the two models wrongly, whereas recall, precision and F1 show the truth.

Other Suitable Measures

Measure	***When to use***
Precision	When false positives are costly (e.g., spam filter blocking important mail).
Recall / sensitivity	When false negatives are costly (e.g., disease screening, fraud).
F1-score	When a balance of precision and recall is needed, especially with imbalanced classes.
ROC-AUC and PR-AUC	Threshold-independent comparison of classifiers; PR-AUC is preferred for highly imbalanced data.
Matthews correlation coefficient, balanced accuracy	Single balanced score for imbalanced binary classification.
MAE, RMSE, R^2	Regression problems.
Silhouette score	Clustering without labels.
Cost-based / business metrics	When the cost of each type of error is known (expected loss, profit).

Q10. Discuss the ethical, social, and technical issues that may arise when deploying Machine Learning systems. Include algorithmic bias, privacy, explainability, security, fairness, accountability, and model drift in your discussion.

When ML systems are used in real life for decisions about people (jobs, loans, health, policing, education), their errors and side effects affect individuals and society. The main issues are as follows.

1. Algorithmic Bias

Bias occurs when a model produces systematically unfair or skewed results because of biased training data, unrepresentative samples, biased labels or design choices. ***Example:*** an experimental recruiting tool developed by a large company was reported to have been dropped after it was found to favour male candidates, because it had learned from past hiring data dominated by men; face

recognition systems have been reported to have higher error rates for darker-skinned women. ***Mitigation:*** representative data, bias audits, re-weighting, diverse teams and testing on subgroups. ***2. Privacy***

ML often needs large amounts of personal and sensitive data (health, location, financial and behavioural data). Risks include unauthorised collection, misuse, data breaches, re-identification of anonymised data and leakage of training data from models (membership inference, model inversion). ***Mitigation:*** consent, data minimisation, anonymisation, encryption, access control, differential privacy, federated learning and compliance with laws such as GDPR and India's Digital Personal Data Protection Act, 2023.

3. Explainability (Transparency)

Many powerful models (deep networks, ensembles) act as "black boxes". If a loan is rejected or a disease is diagnosed, people and regulators need to know why. Lack of explanation reduces trust, makes debugging and detecting bias difficult, and may violate the right to explanation. ***Mitigation:*** use interpretable models where possible, and tools such as feature importance, SHAP and LIME, documentation (model cards) and clear communication.

4. Security

ML systems face special attacks: ***adversarial examples*** (small changes in input that fool the model), ***data poisoning*** (corrupting training data), ***model theft*** and ***evasion*** by fraudsters or malware authors. ***Mitigation:*** adversarial training, input validation, secure pipelines, access control, monitoring and regular security testing.

5. Fairness

Fairness means that the system treats individuals and groups equitably and does not discriminate on the basis of gender, caste, religion, race, age or income. It is difficult because there are several formal definitions (demographic parity, equal opportunity, equalised odds) that cannot all be satisfied together, and removing a sensitive attribute is not enough as other features may act as proxies. ***Mitigation:*** define fairness goals with stakeholders, measure group-wise performance, and apply fairness-aware training or post-processing.

6. Accountability

When an automated decision causes harm, it must be clear who is responsible: the developer, the data provider, the deploying organisation or the user. Lack of accountability leads to harm without remedy. ***Mitigation:*** clear ownership and governance, documentation and audit trails, human-in-the-loop for high-stakes decisions, the right to appeal, impact assessments and compliance with regulation (such as the EU AI Act).

7. Model Drift

After deployment, the real-world data changes. ***Data drift*** is a change in the distribution of inputs, and ***concept drift*** is a change in the relationship between inputs and outputs. Accuracy therefore degrades silently. ***Example:*** a demand forecasting or credit model trained before a pandemic fails when behaviour changes; fraud patterns change as fraudsters adapt. ***Mitigation:*** continuous monitoring, drift detection, periodic retraining and validation, and fallback rules.

Other Social Issues

- ***Job displacement*** due to automation and need for reskilling.
- ***Misinformation and misuse***, such as deepfakes, automated scams and mass surveillance.
- ***Over-reliance and automation bias:*** people trusting the model blindly.
- ***Digital divide and environmental cost:*** unequal access to AI benefits and high energy use of large model training.

Issue	***Risk***	***Main safeguard***
Bias / fairness	Discrimination against groups.	Representative data, audits, fairness metrics.
Privacy	Misuse or leakage of personal data.	Consent, minimisation, encryption, differential privacy.
Explainability	Unexplained, untrusted decisions.	Interpretable models, SHAP / LIME.
Security	Adversarial, poisoning and theft attacks.	Robust training, secure pipelines.
Accountability	No one responsible for harm.	Governance, audit trails, human oversight.
Model drift	Performance decay after deployment.	Monitoring and retraining.